FRC Sensor CAN · Revision B-2 purchasing BOM

51 order lines · 144 fitted components + 10 shunts · Consumed component estimate **US$46.35**

Dated observations: 2026-09-07/08. *First observed price tier per consumed part, excluding MOQ leftovers. Full order budgets including fabrication, assembly, procurement and freight are in COST.md and cost_estimate.csv. No stock is reserved; LCSC inventory is not JLCPCB inventory.

Use BOM_turnkey.csv for a full factory quotation. Match every MPN and placement; blank JLCPCB catalog codes require global sourcing. Ten shunts are bagged accessories. See FACTORY_ORDER.md for panelization, reflow, polarity and programming requirements.

References	Qty	Value / rating	Manufacturer part number	Preferred source	Observed stock	Unit USD*	Used USD*
F1	1	1.5A FAST	**045101.5MRL** Littelfuse	DigiKey F2579CT-ND	7,870	2.4700	2.47
R4, R13, R30, R31, R32, R33, R34	7	1k / 1% / 0.125 W	**0805W8F1001T5E** UNI-ROYAL	LCSC C17513	26,345,600	0.0042	0.03
R3, R6, R9, R11, R14, R15, R21, R22, R24, R25, R26, R40, R41, R105, R205, R305	16	10k / 1% / 0.125 W	**0805W8F1002T5E** UNI-ROYAL	LCSC C17414	47,196,100	0.0039	0.06
R1, R7, R12, R23, R106, R206, R306	7	100k / 1% / 0.125 W	**0805W8F1003T5E** UNI-ROYAL	LCSC C149504	3,765,700	0.0060	0.04
R8, R110, R111, R210, R211, R310, R311	7	4.7k / 1% / 0.125 W	**0805W8F4701T5E** UNI-ROYAL	LCSC C17673	3,907,600	0.0051	0.04
J1, J2, J3	3	2P PUSH 3.5mm	**1824420000** Weidmuller	DigiKey 281-2020-ND	12,805	1.5400	4.62
J4, J5, J6	3	3P PUSH 3.5mm	**1824430000** Weidmuller	DigiKey 281-2021-ND	1,698	2.2100	6.63
Y1	1	8MHz 18pF	**ABM3B-8.000MHZ-B2-T** Abracon	DigiKey 535-9720-1-ND	70,541	0.7600	0.76
Q1	1	AO4485	**AO4485** Alpha & Omega Semiconductor	LCSC C51499	88,125	0.3171	0.32
U2	1	AP2112K-3.3	**AP2112K-3.3TRG1** Diodes Incorporated	DigiKey AP2112K-3.3TRG1DICT-ND	159,040	0.2800	0.28
D102, D103, D202, D203, D302, D303	6	BAT46W	**BAT46W-E3-18** Vishay	LCSC C5557780	4,735	0.1307	0.78
C25, C26	2	30pF / 50V / 5%	**C0805C300J5GACTU** KEMET	DigiKey 399-C0805C300J5GACTUCT-ND	9,211	0.3300	0.66

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C101, C201, C301	3	1nF / 50V / 5%	**CC0805JRNPO9BN102** YAGEO	DigiKey 311-1122-1-ND	105,431	0.1500	0.45
C7, C10, C23, C42	4	1uF / 50V / 10%	**CC0805KKX7R9BB105** YAGEO	DigiKey 311-1886-1-ND	112,870	0.3200	1.28
C28, C29	2	10nF / 50V / 10%	**CC0805KRX7R9BB103** YAGEO	DigiKey 311-1136-1-ND	360,198	0.1100	0.22
C14	1	33nF / 50V / 10%	**CC0805KRX7R9BB333** YAGEO	DigiKey 311-1374-1-ND	24,641	0.1400	0.14
C3, C6, C9, C11, C20, C21, C22, C27, C40, C41, C50, C17	12	100nF / 100V / 10%	**CL21B104KCFNNNE** Samsung Electro-Mechanics	LCSC C28233	605,350	0.0411	0.49
C4, C5, C13, C12, C15, C16	6	22uF / 25V / 10%	**CL31A226KAHNNNE** Samsung Electro-Mechanics	LCSC C12891	211,520	0.1709	1.03
C2	1	10uF / 50V / 10%	**CL32B106KBJNNWE** Samsung Electro-Mechanics	DigiKey 1276-3388-1-ND	149,003	0.6200	0.62
C1	1	47uF / 50V / 20%	**EEE-FK1H470P** Panasonic	DigiKey PCE3815CT-ND	10,108	0.9300	0.93
R2, R107, R207, R307	4	17.8k / 1% / 0.125 W	**ERJ-6ENF1782V** Panasonic Industry	DigiKey P17.8KCCT-ND	33,381	0.1200	0.48
D20	1	ESD2CAN24-Q1	**ESD2CAN24DBZRQ1** Texas Instruments	DigiKey 296-ESD2CAN24DBZRQ1CT-ND	255,809	0.8700	0.87
U7	1	LM339LV	**LM339LVDR** Texas Instruments	DigiKey 296-LM339LVDRCT-ND	1,085	0.4200	0.42
U1	1	LMR51420XDDCR	**LMR51420XDDCR** Texas Instruments	LCSC C5246146	109	1.0292	1.03
D10, D12, D13, D14	4	GREEN	**LTST-C170KGKT** Lite-On	DigiKey 160-1414-1-ND	498,450	0.1900	0.76
D11	1	RED	**LTST-C170KRKT** Lite-On	DigiKey 160-1415-1-ND	489,805	0.1800	0.18
10 loose configuration shunts	10	2 mm shunt	**M22-1900005** Harwin	DigiKey 952-1305-ND	31,908	0.3500	3.50
JP2, JP102, JP202, JP302	4	SELECT	**M22-2010205** Harwin	DigiKey 952-2273-ND	7,644	0.1600	0.64

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JP101, JP201, JP301	3	SELECT	**M22-2010305** Harwin	DigiKey 952-2274-ND	35,632	0.2200	0.66
JP103, JP203, JP303	3	SELECT	**M22-2020305** Harwin	DigiKey 952-2133-ND	838	0.4100	1.23
D3	1	12V ZENER	**MMSZ12T1G** onsemi	DigiKey MMSZ12T1GOSCT-ND	34,429	0.1700	0.17
R101, R102, R201, R202, R301, R302	6	110k / 1% / 0.125 W	**RC0805FR-07110KL** YAGEO	DigiKey 311-110KCRCT-ND	5,943	0.1000	0.60
R103, R203, R303	3	20k / 1% / 0.125 W	**RC0805FR-0720KL** YAGEO	DigiKey 311-20.0KCRCT-ND	27,443	0.1000	0.30
R5	1	3.9k / 1% / 0.125 W	**RC0805FR-073K9L** YAGEO	DigiKey 311-3.90KCRCT-ND	131,203	0.1000	0.10
R10	1	59k / 1% / 0.125 W	**RC0805FR-0759KL** YAGEO	DigiKey 311-59.0KCRCT-ND	29,786	0.1000	0.10
R108, R208, R308	3	7.15k / 1% / 0.125 W	**RC0805FR-077K15L** YAGEO	DigiKey 311-7.15KCRCT-ND	48,097	0.1000	0.30
R104, R204, R304	3	910k / 1% / 0.125 W	**RC0805FR-07910KL** YAGEO	DigiKey YAG3336CT-ND	58,915	0.1000	0.30
R20	1	0 / jumper / 0.125 W	**RC0805JR-070RL** YAGEO	DigiKey 311-0.0ARCT-ND	1,733,368	0.1000	0.10
R42	1	120 / 1% / 0.25 W	**RC1206FR-07120RL** YAGEO	DigiKey 311-120FRCT-ND	126,570	0.1000	0.10
R109, R209, R309	3	4.7k / 1% / 0.75 W	**RC2010FK-074K7L** YAGEO	DigiKey 13-RC2010FK-074K7LCT-ND	7,566	0.2300	0.69
D101, D201, D301	3	SMAJ30A	**SMAJ30A** Littelfuse	LCSC C148230	30,185	0.1459	0.44
D2	1	SMBJ22A	**SMBJ22A-E3/52** Vishay	DigiKey SMBJ22A-E3/52GICT-ND	705	0.5600	0.56
L2	1	4.7uH	**SRP4020TA-4R7M** Bourns	DigiKey SRP4020TA-4R7MCT-ND	9,132	1.0700	1.07
L1	1	10uH	**SRP7028A-100M** Bourns	DigiKey SRP7028A-100MCT-ND	9,366	1.1000	1.10

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D4	1	SS14	**SS14-E3/61T** Vishay	DigiKey SS14-E3/61TGICT-ND	99,297	0.4700	0.47
U5	1	STM32F042K6T6	**STM32F042K6T6** STMicroelectronics	DigiKey 497-14647-ND	3,072	4.3400	4.34
U6	1	TCAN3413DR	**TCAN3413DR** Texas Instruments	LCSC C22433320	233	1.0227	1.02
C8, C24	2	4.7uF / 25V / 10%	**TMK212AB7475KG-T** Taiyo Yuden	DigiKey 587-2990-1-ND	718,393	0.2600	0.52
U3	1	TPS1H200A-Q1	**TPS1H200AQDGNRQ1** Texas Instruments	DigiKey 296-50416-1-ND	25,656	1.7800	1.78
U4	1	TPS2553DBVR-1	**TPS2553DBVR-1** Texas Instruments	LCSC C111738	12,065	0.2978	0.30
U8	1	TPS613222A	**TPS613222ADBVR** Texas Instruments	LCSC C2071163	3,545	0.3697	0.37